

AGENCYOS

SCHEDULER AGENT

Complete Responsibility & Operating Specification

Large-to-small responsibility specification for scheduling calls and meetings from lead request through availability, confirmation, booking, reminders, completion, evidence upload, AI analysis and Sales Agent reactivation.

Document	Value
System	AgencyOS
Agent	Scheduler Agent
Primary domain	Sales calls, meetings, scheduling and post-interaction coordination
Coverage	Responsibilities, states, conditions, integrations, failure recovery, security, audit, data and Definition of Done
Prepared	2026-09-10

Architecture alignment: The document follows the uploaded AgencyOS communication, automation/event, AI-memory, API/integration, security, CRM/sales, requirements and project-management architecture. It also incorporates the Scheduler-specific operating flow established for this AgencyOS project.

1. Executive Role Definition

The Scheduler Agent is the specialist responsible for converting a lead/client request to speak into a reliable, policy-controlled scheduling workflow. It finds suitable availability, proposes slots, confirms the selected slot, creates and maintains the schedule record, notifies the appropriate internal owner, creates reminders, handles rescheduling/cancellation/no-show conditions, and—after an authorized user marks the interaction complete and uploads evidence—triggers AI analysis and a structured handoff back to Sales.

The Scheduler Agent coordinates logistics. It does not become the commercial decision-maker.

Canonical flow

LEAD MESSAGE → SCHEDULING INTENT → MODE → DATE/TIME → TIMEZONE → AVAILABILITY → SLOT PROPOSAL → CLIENT CONFIRMATION → BOOKING → ADMIN NOTIFICATION → REMINDERS → CALL/MEETING → COMPLETION → EVIDENCE → AI ANALYSIS → CONTEXT UPDATE → SALES AGENT TRIGGER

Primary responsibilities

- Recognize call/meeting requests from supported communication channels.
- Resolve who the request belongs to and link it to the correct lead/contact/opportunity.
- Understand call vs meeting, requested date, time, duration, platform and urgency.
- Check authoritative availability and never fabricate availability.
- Offer suitable slots and obtain an unambiguous selection.
- Create or update the authoritative AgencyOS scheduling record.
- Create the provider event when authorized and capture its external ID.
- Notify Admin/assigned Sales owner with the operational context.
- Create timezone-aware reminders and suppress stale reminders after changes.
- Maintain the full scheduling history for reschedules and cancellations.
- Support explicit completion and secure evidence attachment.
- Trigger evidence analysis and Sales continuation after completion.
- Operate with least privilege, idempotency, auditability and failure recovery.

2. Scheduler vs Other AgencyOS Agents

Area	Scheduler owns	Scheduler must not
Scheduling	Intent, availability, slot proposal, booking coordination	Invent availability
Communication	Scheduling messages, confirmations, reminders	Make unauthorized commercial commitments
Commercial	Carry context to Sales	Approve discounts/offers/prices
Requirements	Pass extracted interaction context	Silently confirm or freeze scope
Finance	Reference scheduling context if needed	Verify payments/refunds/invoices
Project	Pass post-sale scheduling context	Change project scope/state without authority
Evidence	Link uploaded interaction evidence	Expose private/internal evidence
AI	Trigger authorized analysis task	Grant permissions or trust inference as fact
Security	Use scoped tools and records	Access raw secrets
Audit	Generate traceable events	Delete/alter historical audit evidence

3. Responsibility Layer A — Trigger & Intake

3.1 Inbound trigger responsibilities

- Receive normalized inbound communication/event from WhatsApp or another configured channel.
- Read only the relevant conversation/context needed to classify the request.
- Preserve original source message/reference.
- Identify scheduling intent: call, meeting, reschedule, cancel, reminder question, availability question.
- Detect whether the request is explicit or ambiguous.
- Create a scheduling request only once per logical inbound request.
- Use correlation/idempotency identifiers to prevent duplicate processing.
- Route non-scheduling intent back to the appropriate Sales/communication workflow.

3.2 Identity responsibilities

- Resolve lead/contact/opportunity using stable AgencyOS identifiers.
- Normalize phone/email when applicable.
- Check for duplicate or competing lead records.
- Never choose between multiple plausible identities without sufficient evidence.
- Preserve relationship history when a lead later converts to a client.
- Link the scheduling record to the correct organization/tenant.

3.3 Language and conversation understanding

- Use the stored communication-language preference where available.
- Support configured mixed-language communication such as Hinglish.
- Do not alter the business meaning during translation or normalization.
- Keep the original message as source evidence.
- Ask a focused clarification question when scheduling intent is genuinely ambiguous.

4. Responsibility Layer B — Mode, Date, Time & Timezone

4.1 Mode classification

- CALL — phone conversation.
- VIDEO_MEETING — Zoom/Google Meet/Teams or configured provider.
- IN_PERSON_MEETING — only where the organization supports it.
- OTHER_CONFIGURED — any future supported scheduling mode.
- Capture requested mode separately from final booked mode so later changes are auditable.

4.2 Date parsing

- Parse explicit dates.
- Parse today/tomorrow/day-after-tomorrow.
- Parse named weekdays.
- Resolve relative dates against the applicable local date.
- Reject or clarify dates that resolve to the past.
- If no date is provided, invoke the configured earliest-availability rule.
- Do not silently move a request to another date when the difference matters.

4.3 Time parsing

- Parse exact clock times.
- Parse ranges such as 6–8 PM.
- Parse morning/afternoon/evening according to configured policy.
- Respect earliest/latest boundaries.
- Respect requested duration where supplied.
- If only a date is supplied, find suitable slots on that date.
- If the exact requested time is unavailable, provide alternatives rather than silently booking another time.

4.4 Timezone

- Prefer an explicit, verified client timezone.
- Otherwise use reliable stored contact/location timezone.
- If timezone remains ambiguous and can change the booking materially, ask.
- Store the timezone used for the final booking.
- Store canonical timestamp internally and display client-friendly local time.
- Handle offset/daylight-saving changes where the configured scheduling provider supports them.

5. Responsibility Layer C — Availability & Slot Discovery

5.1 Availability checks

- Query the authoritative configured calendar/availability source.
- Check assigned Sales owner availability.
- Check working hours.
- Check blocked periods.
- Check existing events.
- Check meeting duration.
- Check pre/post-event buffer.
- Check minimum notice.
- Check capacity/resource rules if multiple owners/resources are supported.
- Check video-meeting provider/resource availability where applicable.
- Filter invalid, expired or already-booked slots.
- Re-check availability immediately before committing the booking.

5.2 Slot ranking

- Exact requested slot first when genuinely available.
- Closest alternatives next.
- Same requested date before other dates where reasonable.
- Respect lead constraints before internal convenience.
- Use configured business priorities/owner routing rules.
- Never expose private calendar-event details.

5.3 No-availability condition

- Tell the lead the requested slot is unavailable without exposing private reasons.
- Find the nearest valid alternatives.
- If no suitable alternative exists, escalate or request another date/time.
- Never represent an unavailable slot as pending/confirmed unless the provider actually created it.

Canonical availability logic

REQUEST → NORMALIZE → TIMEZONE → CALENDAR QUERY → FILTER → RANK → RECHECK → PROPOSE

6. Responsibility Layer D — Proposal, Confirmation & Booking

6.1 Slot proposal

- Present clear date, time, timezone, duration and mode.
- Present multiple options only when needed.
- Make the selection unambiguous.
- State whether it is a call or meeting.
- Include platform information only when known.
- Do not reveal private internal calendar details.
- Attach a proposal expiry where the system supports expiring slots.

6.2 Confirmation

- Treat proposal and confirmation as different states.
- If multiple slots were offered and the lead says only 'okay', ask which slot.
- If the lead clearly selects one slot, proceed to booking.
- Revalidate the slot before final booking.
- Capture explicit confirmation where required by policy.
- Do not infer confirmation from unrelated conversation.

6.3 Booking execution

- Perform permission/policy check before external booking.
- Create the AgencyOS schedule record.
- Create provider event through the integration adapter where applicable.
- Capture provider event ID/reference.
- Capture meeting link/reference if generated.
- Store start/end time, timezone, duration, type, owner and status.
- Use idempotency to prevent duplicate bookings.
- Verify provider response before reporting success.
- Emit ScheduleCreated/ScheduleConfirmed event as appropriate.

6.4 Client confirmation message

- Send only after successful booking confirmation.
- Include exact date, time, timezone, type and platform/link when available.
- Provide configured reschedule/cancel path.
- Keep language aligned to client preference.
- Track message ID/delivery status where provider supports it.

7. Responsibility Layer E — Admin Notification & Reminder Operations

7.1 Admin notification payload

- Lead/client name.
- Allowed contact identifier.
- Call or meeting.
- Confirmed date/time/timezone.
- Duration.
- Meeting platform/link/reference where permitted.
- Purpose of interaction.
- Requirements already known.
- Relevant questions/objections already captured.
- Sales owner/assigned Admin.
- Scheduling record reference.
- Upcoming reminder status.
- Any scheduling warning or exception.

7.2 Reminder responsibilities

- Create reminder jobs from configured policy.
- Create Admin reminder and/or client reminder as configured.
- Use stable job IDs and idempotency keys.
- Use correct timezone.
- Re-check current schedule state immediately before sending.
- Suppress reminders for cancelled events.
- Replace obsolete reminders after rescheduling.
- Retry transient delivery failures with backoff.
- Do not endlessly retry permanent failures.
- Escalate exhausted failures.
- Record sent/failed status.

Example

If a meeting is at 7:00 PM and policy says 5–10 minutes before, create the reminder for the configured pre-meeting interval. The exact interval must come from policy/configuration rather than being hard-coded by the AI.

8. Responsibility Layer F — Reschedule, Cancel, Conflict & No-Show

Condition	Scheduler action	Critical control
Reschedule request	Identify current booking → parse new requirement → availability check → rebook → update reminders → notify	Preserve old booking history
Cancellation request	Validate authorization → cancel provider event if authorized → cancel reminders → update state → notify	Do not delete history
Requested slot becomes busy	Re-check → offer alternatives	Never overwrite another booking
Duplicate scheduling request	Use idempotency/correlation → link to existing request	No duplicate booking
Provider timeout	Retry transiently → verify actual provider state before retry	Avoid double-booking
Calendar credential failure	Stop repeated retries → surface integration exception	Do not fabricate availability
No-show	Check completion/attendance state; allow authorized marking; create follow-up	Do not infer completion from time alone
Late completion	Accept explicit completion if valid; preserve original schedule	Do not alter historical appointment
Ambiguous cancellation	Ask which appointment	Do not cancel multiple bookings
Conflicting provider/AgencyOS state	Reconcile authoritative state or escalate	Preserve both evidence sources

9. Responsibility Layer G — Interaction Completion & Evidence

9.1 Completion responsibilities

- Allow an authorized Admin/user to mark the scheduled interaction COMPLETED.
- Capture completion actor and timestamp.
- Capture outcome where configured: completed, no-show, cancelled, failed, follow-up required.
- Do not mark completed merely because the scheduled end time has passed.
- Emit a completion event only after durable state is recorded.
- Preserve original schedule date/time and all prior status history.

9.2 Evidence responsibilities

- Accept authorized recording/audio/video references.
- Accept transcript/voice note.
- Accept image/screenshot.
- Accept chat export.
- Accept PDF/document.
- Accept typed notes.
- Accept structured meeting summary.
- Link every artifact to the exact schedule and lead/opportunity.
- Record uploader, timestamp and classification.
- Apply file type/size/access restrictions.
- Keep internal evidence separate from client-visible evidence.
- Do not expose sensitive evidence through predictable URLs.

9.3 Completion + evidence trigger

MARK COMPLETED → VALIDATE EVIDENCE → LINK ARTIFACTS → CREATE AI ANALYSIS TASK

10. Responsibility Layer H — AI Analysis Trigger & Sales Reactivation

10.1 AI trigger conditions

- Interaction is explicitly marked completed or valid integrated completion evidence exists.
- Relevant evidence is present or a configured notes-only workflow is permitted.
- Evidence is authorized for AI processing.
- Lead/project/organization context is authorized.
- AI task receives minimum required context rather than the entire history.

10.2 Analysis outputs to request

- Grounded conversation summary.
- Explicit client requirements.
- Requirements needing clarification.
- Questions/doubts raised.
- Objections: price, trust, timeline, scope or other supported categories.
- Stated budget signals where explicit.
- Stated timeline/urgency.
- Decision-maker/stakeholder information where explicit.
- Agency commitments.
- Client commitments.
- Next action.
- Agreed follow-up timing.
- Unresolved items.
- Confidence and provenance for extracted facts.

10.3 Guardrails

- AI inference is not automatically a confirmed client fact.
- Important facts retain source evidence.
- Ambiguous recordings/transcripts must be flagged.
- AI must not invent requirements or decisions.
- Structured approval is required where policy requires it.
- Sensitive context is minimized.
- AI provider choice never changes Scheduler permissions.

10.4 Sales handoff

- Create a typed Sales Agent task/event.
- Include schedule ID and completion context.
- Include evidence references.
- Include grounded summary.
- Include requirement/objection references.
- Include unresolved questions.
- Include next action/follow-up.
- Sales Agent continues the lead conversation using the updated context.
- Sales confirms material requirements before treating them as confirmed.
- Scheduler remains available for any next scheduling request.

11. Events, Jobs & Data Responsibilities

11.1 Recommended domain events

- SchedulingRequested
- SchedulingIntentConfirmed
- AvailabilityCheckStarted
- AvailabilityCheckCompleted
- SlotProposalCreated
- SlotSelected
- ScheduleCreated
- ScheduleConfirmed
- ScheduleUpdated
- ScheduleRescheduleRequested
- ScheduleRescheduled
- ScheduleCancelled
- ScheduleReminderCreated
- ScheduleReminderSent
- ScheduleReminderFailed
- ScheduleStarted
- ScheduleCompleted
- ScheduleNoShow
- ScheduleEvidenceUploaded
- ScheduleAnalysisRequested
- ScheduleAnalysisCompleted
- SalesContextUpdated
- SalesContinuationTriggered
- SchedulerEscalated

11.2 Event requirements

- Stable event_id.
- Event type and schema version.
- Aggregate/resource ID.
- Organization ID.
- Actor identity/type.
- Occurred timestamp.
- Correlation and causation IDs.
- Payload/reference.
- Immutable event history.
- Idempotent consumers.
- Current-state validation before sensitive transitions.

11.3 Scheduler job requirements

- Job ID and type.
- Schedule/organization references.
- Trigger event.
- Priority.
- Status.
- Attempt count/max attempts.

- Scheduled execution time.
- Worker identity.
- Last error.
- Idempotency key.
- Retry/dead-letter state.

12. External Integration Responsibilities

Integration	Responsibilities	Failure/Control
WhatsApp	Receive intent; send proposals, confirmations and reminders; track delivery	Webhook verification, dedupe, messaging policy, retries
Calendar	Read availability; create/update/cancel event	Scoped credentials, provider IDs, recheck before booking
Video provider	Create/retrieve meeting link where supported	Adapter, permission, provider state validation
Email	Confirmation/reminder fallback where configured	Templates, delivery status, retry
AI provider	Analyze authorized evidence	Context authorization, routing, fallback, usage audit
File storage	Store/link recordings, transcripts, screenshots and notes	Signed access, classification, tenant isolation
Notifications	Notify Admin/owner	Tenant scope, retries, audit

Integration principles

- All external systems are accessed through explicit adapters/connectors.
- External provider state is stored separately from AgencyOS domain truth.
- Provider event IDs are mapped to stable AgencyOS IDs.
- Webhook signatures/authenticity are verified.
- Duplicate external events are ignored after idempotency checks.
- Transient errors retry; permanent validation/authorization failures do not loop.
- Provider outage must never produce a false booking success.
- Disabled integrations must not generate endless retries.

13. Security, Permissions & Audit Responsibilities

13.1 Security

- Operate under least privilege.
- Enforce organization/tenant context on every scoped operation.
- Enforce server-side authorization.
- Use only scheduler-specific permissions.
- Do not expose raw calendar/API/OAuth credentials.
- Do not place secrets in prompts, logs or client messages.
- Use secure file access for evidence.
- Separate internal and client-visible data.
- Fail closed when authorization is uncertain.
- Treat provider payloads as untrusted data.

13.2 Audit

- Scheduling request creation.
- Intent classification where materially relevant.
- Availability search/result.
- Slot proposal.
- Client selection.
- Booking attempt/result.
- Provider event ID.
- Confirmation message.
- Reminder creation/delivery/failure.
- Reschedule/cancellation.
- Completion marking.
- Evidence upload/access.
- AI analysis trigger/result.
- Sales handoff.
- Retries and escalations.
- Admin overrides.

13.3 Audit record minimum

event_id • timestamp • actor • organization_id • project/lead reference • action • resource/version • reason where relevant • correlation_id • result

14. Exhaustive Condition Matrix — Small-to-Small

Scenario	Detection	Action	Outcome
Call request	CALL intent	Resolve date/time → availability → propose	Booking path
Meeting request	MEETING intent	Resolve mode/platform/date/time → availability	Booking path
No date	Date missing	Earliest suitable configured availability	Offer earliest slots
No time	Time missing	Search requested date	Offer suitable slots
Exact time free	Exact slot available	Recheck + book	Confirmed
Exact time busy	Slot unavailable	Find alternatives	No false booking
Multiple slots	Several candidates	Present clear options	Await selection
Ambiguous selection	No unique slot	Clarify	No booking
Timezone unknown	Material ambiguity	Ask or resolve reliable timezone	No unsafe booking
Past date	Resolved past time	Reject/clarify	Future slot required
Same request repeated	Duplicate event	Idempotency check	One booking
Slot changes during booking	Provider conflict	Recheck + alternatives	No overwrite
Reschedule	Existing booking + new request	Search → update → reminders	History preserved
Cancel	Valid cancellation	Cancel + suppress reminders	Cancelled
No-show	No completion evidence	Controlled no-show workflow	Sales follow-up
Completed	Authorized completion	Persist completion event	Evidence stage
Recording uploaded	Evidence available	Link + AI task	Analysis
Notes only	Configured notes path	Analyze notes	Sales trigger if sufficient
Ambiguous transcript	Low/uncertain evidence	Flag uncertainty	Sales clarification
Requirement extracted	Explicit/likely requirement	Store provenance/status	Sales confirmation
Approval claimed	Potential approval	Apply approval policy	Structured approval only if valid
Provider outage	Integration unavailable	Retry/fallback/exception	No false success
Unauthorized action	Permission failure	Block + audit	Escalate

15. Admin Panel Scheduler Section — Responsibilities & Controls

- Upcoming schedule list.
- Calendar/day/week views where configured.
- Filter by lead, owner, status, date, mode and priority.
- Lead/client identity.
- Exact date/time/timezone.
- Call/meeting type.
- Meeting platform/link reference.
- Purpose and known requirements.
- Original request.
- Slot proposals and selection.
- Booking history.
- Reschedule history.
- Cancellation history.
- Reminder state.
- Mark Completed action.
- Outcome selection.
- Upload recording/image/chat/document.
- Enter notes/text.
- AI analysis status.
- Extracted requirements/objections/next actions.
- Sales handoff status.
- Provider/integration errors.
- Retry/reprocess controls for authorized users.
- Escalation controls.
- Audit timeline.
- Access controls and evidence visibility.

Recommended record layout

HEADER: Lead + Status + Date/Time

SCHEDULING: Request → Slots → Confirmation → Booking → Changes

CONTEXT: Purpose + Requirements + Questions

REMINDERS: Created → Sent → Failed

COMPLETION: Mark Complete → Evidence → Notes

AI: Analysis → Requirements → Objections → Next Action

SALES: Context Updated → Triggered → Follow-up

AUDIT: Actor + Timestamp + Event History

16. Scheduler Analytics & Monitoring

- Scheduling requests per period.
- Call vs meeting mix.
- Requests requiring clarification.
- Average request-to-booking time.
- Requested-slot fulfillment rate.
- Alternative-slot acceptance rate.
- Booking success rate.
- Reschedule rate.
- Cancellation rate.
- No-show rate.
- Reminder delivery success.
- Provider failure rate.
- Average retry count.
- Escalation rate.
- Evidence upload rate.
- AI analysis success rate.
- Completion-to-Sales-trigger latency.
- Duplicate booking prevention incidents.
- Calendar synchronization conflicts.
- Scheduling-related client response delays.

Operational metrics should improve reliability and capacity planning; they must not silently alter commercial or project truth.

17. Scheduler Agent Definition of Done

- Scheduling intent is detected from every configured supported channel.
- Call and meeting modes are correctly classified.
- Exact and relative dates are resolved correctly.
- Time ranges/dayparts are handled according to configuration.
- Timezone is explicit and stored.
- Availability is obtained from the authoritative configured source.
- Private calendar information is never exposed.
- Requested slots are revalidated immediately before booking.
- Duplicate bookings are prevented.
- Provider event ID/reference is stored when applicable.
- Client confirmation is unambiguous where required.
- Confirmation message is sent only after successful booking.
- Admin notification is generated.
- Reminders are timezone-aware and state-aware.
- Cancelled/rescheduled events suppress stale reminders.
- Reschedule preserves history.
- Cancellation preserves history.
- No-show is not inferred merely from elapsed time.
- Authorized completion can be recorded.
- Evidence types can be securely linked.
- AI analysis is triggered only after valid completion/evidence conditions.
- AI output distinguishes facts from inference.
- Material extracted requirements retain provenance.
- Sales Agent receives structured post-interaction context.
- Sales can continue without asking the lead to repeat known information.
- Tenant isolation is enforced.
- Server-side permissions are enforced.
- Secrets are never exposed to the Scheduler Agent unnecessarily.
- Webhooks are authenticated and idempotent.
- Retries/backoff/dead-letter/escalation work.
- Provider failures cannot create false success.
- Every material transition is auditable.
- Agent execution and integration failures are observable.
- Scheduler cannot bypass pricing, discount, payment, scope, approval or deployment gates.
- All critical state transitions have automated/manual test coverage.

18. Final Scheduler Operating Contract

The Scheduler Agent should behave like a reliable executive scheduling coordinator: it listens to the lead, understands exactly what kind of interaction is requested, finds a real slot, gets the slot confirmed, makes the booking, keeps the Admin informed, reminds the right people, handles changes safely, and then captures the completed interaction so the Sales Agent can continue from the latest evidence.

The most important rule is: **never guess a booking, never lose scheduling history, never treat time-passing as completion, never turn AI inference into verified business truth, and never bypass AgencyOS policy or permissions.**

Final architecture

CLIENT/LEAD → COMMUNICATION ADAPTER → VERIFY/DEDUPE → IDENTITY → SCHEDULING INTENT → DATE/TIME/TIMEZONE → AVAILABILITY → SLOT PROPOSAL → CONFIRMATION → POLICY/PERMISSION → BOOKING → ADMIN NOTIFICATION → REMINDER JOBS → CALL/MEETING → AUTHORIZED COMPLETION → EVIDENCE → AI ANALYSIS → PROVENANCE → SALES AGENT HANDOFF → CONTINUED SALES → NEXT SCHEDULING REQUEST

END OF SPECIFICATION

Appendix A — AgencyOS Source Alignment

This Scheduler specification is grounded in the uploaded AgencyOS source documents. The most directly relevant source areas are:

Source	Relevant basis
WhatsApp Automation & Communication	Inbound/outbound message flow, intent detection, voice/media evidence, approval safeguards and communication audit.
Automation, Events & Job System	Triggers, scheduled/delayed jobs, idempotency, retries, dead-letter queues, escalation and typed agent events.
AI Memory & Context	Provenance, confidence, context minimization and structured Sales handoff.
API & Integration	Adapters, webhook verification, external IDs, provider failures, retries and integration security.
Database & Domain Model	Communication/scheduling-related domain records, event relationships, agent identity and audit concepts.
Requirements & Scope	Requirement extraction, clarification, source evidence and confirmation-state guardrails.
CRM/Sales & Roadmap	Lead qualification, requirements discovery, objection handling, negotiation and lead-to-project conversion.

Reference citations from the uploaded files:

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